SABAHAT IJAZ

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▷ Experience

Senior Software Engineer — **Python, AI**
Quillio
March 2025 – Present
Building an AI legal assistant that helps lawyers with document review, research, drafting, and client intake. Designed a multi-provider LLM backend with retrieval-augmented chat over case files, multi-document analysis, live progress updates, and voice input and output.

Senior Software Engineer — **Team Lead**
Expert System Solution
March 2023 – March 2025
Led Python backend and generative AI delivery for remote clients in the USA, Los Angeles, and Turkey. Shipped production image-generation and virtual try-on systems, and mentored engineers on model integration, APIs, and release quality.

Software Engineer — **Python, Backend**
Reclosure (ProptoXchange)
March 2024 – February 2025
Built the multi-tenant API for ProptoXchange, Reclosure's platform for tokenizing and trading real-world assets. Owned authentication and role-based access, tenant-scoped marketplace and property APIs, KYC/KYB integrations, document workflows, compliance and ledger jobs, and the PostgreSQL data layer, including XRPL integrations.

Team Lead and Data Scientist — **Team Lead**
Odyssey Solutions
August 2022 – March 2023
Led Y-Hat, the company's automated time-series forecasting product. The tool ingests data and runs preprocessing, feature engineering, feature selection, and model training, then deploys the best-performing model in one click.

Data Science Engineer — **Data Scientist**
Odyssey Solutions
November 2021 – July 2022
Built product features and proofs of concept, including cryptocurrency-price and power-price forecasts. Automated the machine-learning workflow for classification, regression, and time series.

Machine Learning Engineer — **ML Engineer**
Ecologix US
July 2020 – October 2021
Worked on internal products and outsourced projects: distributed computing and distributed file storage, time-series prediction, Numenta Hierarchical Temporal Memory, an eBay category hierarchy in Neo4j, and live video-stream analysis.

▷ Skills

Generative AI and LLMs — OpenAI, Anthropic Claude, Google Gemini, AWS Bedrock, and Groq. Prompt design, tool use, structured outputs, streaming responses, and multi-provider routing with cost and load balancing. Speech-to-text and text-to-speech. Diffusion: Stable Diffusion 1.5, 2.0, and SDXL, ControlNet, inpainting, outpainting, text-to-image, image-to-image, and SDXL fine-tuning.

RAG and AI Agents — Retrieval-augmented generation, embeddings, and vector search with Weaviate and Pinecone. Document QA, long-term memory, and multi-agent workflows for research and email tagging and reply. LangChain, LlamaIndex, and Phidata. Tracing with OpenTelemetry and Sentry.

Machine Learning — Classification, regression, time-series forecasting, anomaly detection, natural language processing, and graph ML. LSTM, ARIMA family, Prophet, N-BEATS, N-HiTS, NeuralProphet, Transformers, and hybrid SARIMAX-LSTM. TensorFlow, Keras, scikit-learn, Pandas, and NumPy.

Computer Vision — Object detection, tracking, semantic and instance segmentation, and live video analysis. YOLO, Segment Anything, GroundingDINO, SegFormer, CLIP, Mask R-CNN, Faster R-CNN, and Vision Transformers. OpenCV, Pillow, scikit-image, and Matplotlib.

Backend, Data, and Cloud — Python, FastAPI, Flask, Pydantic, REST, and WebSockets. Celery, Redis, Docker, and Jenkins. PostgreSQL, SQLAlchemy, Alembic, MongoDB, MySQL, and Neo4j. Auth0, JWT, and role-based access. AWS S3. Git and Linux.

Languages — Python, SQL, C/C++, JavaScript, HTML, and CSS.

▷ Projects

AILA – AI Legal Assistant
(March 2025-Present)
Built the backend for an AI-powered legal assistant using FastAPI, integrating multiple LLM providers (OpenAI, Anthropic, Gemini) for document review, legal research, and drafting. Implemented a multi-document analysis system (COMPAS AI) with real-time WebSocket updates and parallel worker processing. Developed RAG-based contextual chat with vector search (Weaviate), long-term memory, and streaming responses. Integrated voice capabilities (speech-to-text/text-to-speech) and an AI-powered legal intake assistant. Engineered a cost-tracking and load-balancing layer across multiple AI providers to optimize performance and spending.
Tools/Technology: Python, FastAPI, MongoDB, Redis, Weaviate, Celery, OpenAI API, Anthropic API, Gemini API, AWS Bedrock, LangChain, LlamaIndex, WebSockets, Docker, Sentry, OpenTelemetry, Stripe, Deepgram, Google Document AI

ProptoXchange
(March 2024–February 2025)
Multi-tenant API for Reclosure's platform for tokenizing and trading real-world assets such as property, private credit, and funds. Built Auth0 authentication and role-based access for superadmins, tenant admins, and asset managers, with tenant-scoped marketplace and property APIs and asset-level memberships. Integrated KYC/KYB providers, S3 presigned document upload and download, and XRPL token flows. Added scheduled jobs for compliance checks, ledger tracking, and data backfills on PostgreSQL with SQLAlchemy and Alembic migrations.
Tools/Technology: Python, FastAPI, Pydantic, SQLAlchemy, PostgreSQL, Alembic, Auth0, JWT, RBAC, AWS S3, APScheduler, XRPL, REST, Docker, Git

Automatic Email-Agent
(January 2025-March 2025)
An email agent that tags incoming mail and drafts replies.
Tools/Technology: Python, LangChain, Phidata, Groq, Pinecone, OpenAI, Streamlit

SWOT-ANALYZER
(January 2025)
A multi-agent chatbot that takes any company, product, or topic and returns a SWOT-style analysis plus its top five competitors.
Tools/Technology: Python, LangChain, Phidata, Groq, Pinecone, OpenAI, Streamlit

pdfQA
(October 2024)
A chatbot for asking questions about an uploaded PDF.
Tools/Technology: Python, Groq, OpenAI, Streamlit

Stylar
(March 2023-September 2024)

A virtual try-on platform where a user selects a product and sees it on themselves. Built on Stable Diffusion, ControlNet, BLIP/BLIP-2, and ChatCaptioner, with outpainting, synthetic data generation, image scraping, and automatic image filtering. Includes an SDXL 1.0 model fine-tuned for shoe styles.
Tools/Technology: Python, Transformers, OpenAI, Stable diffusers, ControlNet, Computer Vision Pandas,Tensorflow, Numpy, Huggingface

GenAI Superficial
(March 2023-September 2024)

A generative-image platform on Stable Diffusion and OpenAI models. Features include BLIP and BLIP-2 captioning, ChatCaptioner, inpainting, automatic background removal and replacement, outpainting, image-to-image, and text-to-image generation.
Tools/Technology: Python, Transformers, OpenAI, Stable diffusers, ControlNet, Computer Vision Pandas,Tensorflow, Numpy, Huggingface

ControlNet-SegmentAnything-GroundingDINO
(March 2023-September 2024)

A platform with the combination of powerful image detectors and generators to enhance data generation process
Tools/Technology: Python, FLASK, Transformers, Computer Vision Pandas,Tensorflow, Numpy

UseCase Specific ChatGPT
(March 2023-September 2024)

Creating a platform where user can modify GPT model according to his needs
Tools/Technology: Python, Transformers,Natural Language Processing, Tensorflow, Numpy

Y-Hat
(November 2021-January 2023)

A platform that automates time-series analysis and forecasting, with modules for preprocessing, feature engineering, feature selection, visualization, and analysis. Models include LSTM, SARIMAX, and hybrid SARIMAX-LSTM.
Tools/Technology: Python, Flask, Flasgger, PostgreSQL, React, ETL, AWS, Jenkins, Pandas, NumPy, Vaex

Power Price POC
(March 2022-July 2022)

Developed POC for Power Price forecasting. Multivariate and Multi-Step Forecast problem. Developed models that forecast price for upto 1 year. Models forecasted next 24 hours power price with more than 90-percent accuracy. Worked on Encoder-Decoder LSTM , Neural Prophet and SARIMAX and more
Tools/Technology: Python, Flask, Flasgger, Pandas, NumPy

Crypto POC
(February 2022–May 2022)

Developed POC for crypto currencies price forecasting. Focused on Bitcoin and Ethereum. Multivariate and Multi-Step Forecast problem. Developed models that forecast price for upto 1 year. Developed Pipeline that gets updated data from online and updated model on daily basis on start of crypto market. Models forecasted upto 1 month with more than 90-percent accuracy. Worked on Encoder-Decoder LSTM and SARIMAX
Tools/Technology: Python, Flask, Flasgger, Pandas, NumPy

Big Data Storage and Analysis
(August 2021-October 2021)

Stored Big data in Hadoop HDFS system and used Spark to query data. Stored Big data through Hive and used Spark to query data. Used Hive with HBase to overcome latency issue. Ingested Data into HDFS through Flume. Used PigLatin to query data.
Tools/Technology: Hadoop (HDFS and MapReduce), Spark, Hive

NiHA
(February 2021-October 2021)

Automated Listing work, Purchase and Sale of Ebay. Automated customer handling. Automated the process of purchasing product from a vendor on Ebay and adding product to owners or clients vendor page. Automated the process of analyzing which product or category is going to catch up the market.
Tools/Technology: Python, C#, CQL, Neo4j, Selenium, BeautifulSoup4

Time Series Prediction and Anomaly Detection
(July 2021-July 2021)

Used HTM model proposed by Numenta for Time Series and Anomaly Prediction. Used this technique for stock market trend prediction and trend prediction on E-Commerce Websites.
Tools/Technology: Machine Learning/Python, Linux, ReactJS, FLASK.

Social Media Sentiment Analysis
(July 2021-August 2021)

Used textBlob, Transformers and LSTM to analyse sentiments in data scrapped from Twitter, Reddit, News and Youtube. Managed all data in Hadoop Ecosystem.
Tools/Technology: Machine Learning/Python, Hadoop,Natural Language Processing, Web-Scrapping, Linux.

eBay Hierarchy Mapping in Neo4j
(February 2021-April 2021)

Scrapped and mapped all the hierarchical data of Ebay in Neo4J Graph Database. Part of Firms Product.
Tools/Technology: Python, C#, CQL, Neo4j

Video Stream Analyzation
(February 2021)

Extracted 4 sides of product from image to automate listing work.
Tools: Used Image processing Techniques for analyzing products similar to title, and detected defects in product using image.

▷ Education

MS (DS) Data Science
2024, FAST NUCES, Lahore.

Master's in Data Science from FAST NUCES – CGPA: 3.00
Major courses: Machine Learning, Deep Learning, Tools and Techniques for Data Science, Statistical and Mathematical Methods for Data Science

BS (CS) Computer Science
2020, FAST NUCES, Lahore.

Bachelor's in Computer Science from FAST NUCES - CGPA: 3.41
Some Major Courses: Data Structures A-, Design & Analysis of Algorithms A+, Computer Networks A, Operating Systems A-, Artificial Intelligence B+, Machine Learning A-, Computer Modeling and Simulation. BS Thesis Title: Pathological Myopia Detection Using Fundus Images

▷ AwardsHonors

Deans of Honor
Honor

National University of Computer & Emerging Sciences
Awarded with Deans of Honor in Fourth, Fifth and Seventh semester

Gold Medal
Award

National University of Computer & Emerging Sciences
Achieved Gold Medal (Summa Cum Laude) in the Fourth semester

Silver Medal
Award

National University of Computer & Emerging Sciences
Achieved Silver Medal (Magna cum Laude) in Fifth Semester and in BS overall

Bronze Medal
Award

National University of Computer & Emerging Sciences
Achieved 4 Bronze (cum laude) Medals in First, Second, Third and Seventh Semester

▷ References